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Modern Communication Systems F25

Lab 8

12/06/2025

Theory:

1. Quadratic Chirp FMCW Radar

a. instantaneous frequency $f(t)$

$$f(t) = \frac{1}{2\pi} \frac{d}{dt} (\phi(t))$$

$$\phi(t) = 2\pi (f_0 t + \frac{k}{3} t^3)$$

$$\frac{d\phi}{dt} = 2\pi (f_0 + k t^2)$$

$$f(t) = f_0 + k t^2$$

b. delayed echo

$$y(t) = a x(t-\tau)$$

$$y(t) = a \exp(j 2\pi (f_0 (t-\tau) + \frac{k}{3} (t-\tau)^3))$$

$$y(t) = a \exp(j 2\pi (f_0 (t-\tau) + \frac{k}{3} t^3 - k t^2 \tau + k t \tau^2 - \frac{k}{3} \tau^3))$$

c. dechirped, derive $A(\tau)$, $p(\tau)$, $q(\tau)$

$$s(t) = a \exp(j 2\pi (f_0 (t-\tau) + \frac{k}{3} (t-\tau)^3)) * \exp(-j 2\pi (f_0 t + \frac{k}{3} t^2))$$

$$s(t) = a \exp(j 2\pi (-f_0 \tau + \frac{k}{3} ((t-\tau)^3 - t^3)))$$

$$s(t) = a \exp(j 2\pi (-f_0 \tau - k \tau t^2 + k \tau^2 t - \frac{k}{3} \tau^3))$$

$$s(t) = a \exp(j 2\pi (-f_0 \tau - \frac{k}{3} \tau^3)) \exp(j 2\pi ((-k \tau) t^2 + (k \tau^2) t))$$

$$A(\tau) = a \exp(j 2\pi (-f_0 \tau - \frac{k}{3} \tau^3)), p(\tau) = -k \tau, q(\tau) = k \tau^2$$

2. FMCW Radar Design

a. required chirp bandwidth

$$\Delta R = c / (2 B_{\text{chirp}})$$

$$B_{\text{chirp}} = c / (2 \Delta R)$$

$$B_{\text{chirp}} = 3 \times 10^8 / (2 \times 0.3)$$

$$B_{\text{chirp}} = 5 \times 10^8 \text{ Hz} = 500 \text{ MHz}$$

b. required chirp duration and slope

$$f_b = 2kR / c$$

$$k = B_{\text{chirp}} / T_{\text{chirp}}$$

$$f_{b\text{max}} = 2kR_{\text{max}}/c \leq f_s/2$$

$$k = f_s c / (4 R_{\text{max}})$$

$$k = 25 \times 10^6 * 3 \times 10^8 / (4 * 150)$$

$$k = 1.25 \times 10^{13} \text{ Hz/s}$$

$$T_{\text{chirp}} = B_{\text{chirp}} / k$$

$$T_{\text{chirp}} = 5 \times 10^8 / 1.25 \times 10^{13} = 4 \times 10^{-5} \text{ s} = 40 \mu\text{s}$$

c. required chirp repetition interval

$$f_D = 2v / \lambda$$

$$\lambda = c / f_{\text{start}}$$

$$f_{D\text{max}} = 1 / (2 T_{\text{rep}})$$

$$1 / (2 T_{\text{rep}}) = 2 v_{\text{max}} / \lambda$$

$$T_{\text{rep}} = \lambda / (4 v_{\text{max}}) = c / (4 f_{\text{start}} v_{\text{max}})$$

$$T_{\text{rep}} = 3 \times 10^8 / (4 * 77 \times 10^9 * 20) = 4.87 \times 10^{-5} \text{ s} = 48.7 \mu\text{s}$$

d. required min chirps in each frame

$$\Delta f_D = 1 / (M T_{\text{rep}})$$

$$\Delta v = \lambda / 2 \Delta f_D = \lambda / (2 M T_{\text{rep}})$$

$$M = \lambda / (2 \Delta v T_{\text{rep}})$$

$$\lambda = c / f_{\text{start}}$$

$$M = (3 \times 10^8 / 77 \times 10^9) / (2 * 0.25 * 4.87 \times 10^{-5}) = 160$$

3. Programming: Breathing and Heart Rate with FMCW Radar

a. coding:

i. Range FFT and range axis

ii. Phase extraction in slow time

iii. Breathing/heart-rate spectrum

b. Theoretical and interpretation questions

i. Using the given FMCW parameters, derive:

1. The range resolution ΔR and compute its numerical value

$$\Delta R = c / (2 B_{\text{chirp}})$$

$$\Delta R = 3 \times 10^8 / (2 * 1 \times 10^9) = 0.15 \text{ m}$$

2. The maximum unambiguous range R_{max} you can observe with the chosen FFT size

$$k = B_{\text{chirp}} / T_{\text{chirp}}$$

$$f_s = N_{\text{sample}} / T_{\text{chirp}}$$

$$f_{\text{max}} = f_s / 2$$

$$R_{\text{max}} = c f_{\text{max}} / (2 k)$$

$$R_{\text{max}} = c f_s / (4 k)$$

$$R_{\text{max}} = c N_{\text{sample}} T_{\text{chirp}} / (4 B_{\text{chirp}} T_{\text{chirp}})$$

$$R_{\text{max}} = c N_{\text{sample}} / (4 B_{\text{chirp}})$$

$$R_{\text{max}} = 3 \times 10^8 * 256 / (4 * 1 \times 10^9) = 19.2 \text{ m}$$

ii. Intuitively explain how the breathing and heart rate of the chest target can be estimated using FMCW signal. No formula needed

each chirp's beat spectrum gives a range profile, so targets at different ranges can be isolated. Small chest movements cause tiny changes in path length, giving tiny changes in the measured phase of the returned signal at that range. If you add across chirps and unwrap the phase, that phase vs time is proportional to chest displacement vs time. Take the FFT of this to get a spectrum where the peaks are breathing rate and heart rate

iii. Using your plots:

1. Report the estimated range (in meters) of the chest target
5m from the first peak in figure 1

2. Report the estimated breathing rate and heart rate (in Hz), as
obtained from the slowtime phase spectrum

Breathing: 0.2Hz from figure 4.

note, the peak in the spectrum is at 0.24Hz due to the low resolution, but with an amplitude of 4.7 at .24 and 2.8 at .12, a simple weighted average of the two peaks gives a decently accurate 0.1953

Heart rate: 1.2Hz with a clear peak

hand motion can also be seen at 5Hz

iv. Discuss the impact of additional motions:

1. What is the effect of the dancing user on the range profile and on
the range-time map?

In figure 1 it shows as a second peak at 20m.

In figure 2 it shows as a second intense line, again at 20m.

2. How does high-frequency limb/hand motion (micro-Doppler)
affect the breathing/heart rate estimation?

It adds another frequency spike in figure 4. With it being so far away at 5Hz it's easily distinguishable, but in certain conditions where these frequencies line up it could cause issues. With relatively large amplitude or transient motion it could also create large phase swings and introduce broadband energy in the slow time spectrum, increasing noise and potentially mask or distort the heart rate.

Plots:



